

8.6

(a)

$$\text{Wage}_i = B_1 + B_2 \text{EDUC}_i + B_3 \text{EXPER}_i + B_4 \text{METRO}_i + \varepsilon_i$$

$$H_0: \sigma_m^2 = \sigma_F^2 \quad \text{vs} \quad H_1: \sigma_m^2 \neq \sigma_F^2$$

$$\text{SSE}_M = 97161.9174 \quad , \quad df_M = 577 - 4 = 573$$

$$S_M^2 = \frac{97161.9174}{573} = 169.567$$

$$S_F^2 = 12.024^2 = 144.577 \quad , \quad df_F = 1000 - 577 - 4 = 419$$

Test statistic

$$F^* = \frac{S_M^2}{S_F^2} \sim F(573, 419)$$

$$RR: \left\{ F^* \mid F^* > F_{0.025}(573, 419) \approx 1.18 \text{ or } F^* < F_{0.975}(573, 419) \right\}$$

$$\therefore F^* = \frac{169.567}{144.577} \approx 1.1728 \notin RR \quad \therefore \text{Do not reject } H_0 \text{ at } \alpha = 0.05$$

There isn't enough evidence to say that $\sigma_m^2 \neq \sigma_F^2$ #

(b)

$$\text{Wage}_i = B_1 + B_2 \text{EDUC}_i + B_3 \text{EXPER} + B_4 \text{METRO}_i + B_5 \text{FEMALE}_i + e_i$$

$$H_0: \sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2 \quad \text{vs} \quad H_1: \sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$$

$$S_S^2 = \frac{\text{SSE}_S}{df_S} = \frac{56231.0382}{400-5} = 142.357, \quad S_M^2 = \frac{\text{SSE}_M}{df_M} = \frac{100703.0471}{600-5} = 169.249$$

Test statistic

$$F^* = \frac{S_M^2}{S_S^2} \sim F(595, 395)$$

$$\text{RR}: \left\{ F^* \mid F^* > F_{0.05}(595, 395) \approx 1.15 \right\}$$

$$\therefore F^* = \frac{169.249}{142.357} \approx 1.1889 \in \text{RR} \quad \therefore \text{Reject } H_0 \text{ at } \alpha = 0.05$$

There is enough evidence to say that $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$ #

(c)

H_0 : model 無異質變異 vs H_1 : model 存在異質變異

Test statistic

$$NR^2 \sim \chi^2(4)$$

$$RR: \{ NR^2 \mid NR^2 > \chi^2_{0.05}(4) \approx 9.488 \}$$

$\therefore NR^2 = 59.03 \in RR \therefore \text{Reject } H_0 \text{ at } \alpha = 0.05$

There is enough evidence to say that the model is heteroskedasticity

雖然 NR^2 檢定是針對所有 independent variable 的整體檢定, 但其證實變異數不是固定的, 這與 (b) 小題算出的變異數隨婚姻狀態改變的觀點有所關聯。

#

(d)

H_0 : model 無異質變異 vs H_1 : model 存在異質變異

White test 中自由度包含原本變數、變數的平方項 (不含 dummy variable) 以及變數的交乘項。

$$df = 4 + 2 + \binom{4}{2} = 12$$

Test statistic

$$NR^2 \sim \chi^2(12)$$

$$RR = \{ NR^2 \mid NR^2 > \chi^2_{0.05}(12) \approx 21.026 \}$$

$\therefore NR^2 = 78.82 \in RR \therefore \text{Reject } H_0 \text{ at } \alpha = 0.05$

There is enough evidence to say that the model is heteroskedasticity.

#

(e)

當 $robuse < se$ 時 估計區間變窄, 故 EXPER、MARRIED、FEMALE 的估計區間變窄

當 $robuse > se$ 時 估計區間變寬, 故 Intercept、EDUC 的估計區間變寬

此結果不存在矛盾, 在存在異質變異的情況下, 普通OLS的標準誤估計值會產生biased, 但這種偏差的方向是不確定的, OLS 標準誤可能會低估真實的標準誤亦可能高估, 穩健標準誤的目的不是為了讓區間變窄或變寬, 而是為了提供一個在異質變異下更一致且可靠的估計。

#

(f)

t value檢定代表MARRIED變數對薪資平均值的解釋力, 當 t值趨近於1 (小於臨界值約 1.96) 時, 代表在統計上我們無法證實已婚者的平均薪資與單身者有顯著不同, 而 (b) 小題則檢定已婚者與單身者的薪資波動程度不同, 此兩種結果是相容的。

#

8.16

(9)

```
> model_ols <- lm(miles ~ income + age + kids, data = vacation)
> summary(model_ols)
```

Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:

Min	1Q	Median	3Q	Max
-1198.14	-295.31	17.98	287.54	1549.41

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-391.548	169.775	-2.306	0.0221 *
income	14.201	1.800	7.889	2.10e-13 ***
age	15.741	3.757	4.189	4.23e-05 ***
kids	-81.826	27.130	-3.016	0.0029 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

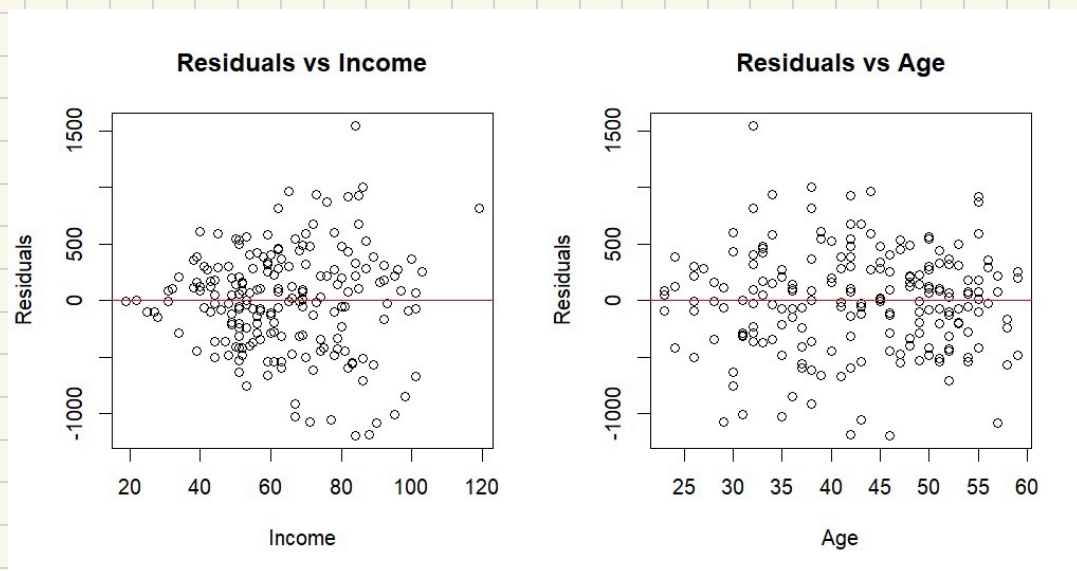
Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared: 0.3406, Adjusted R-squared: 0.3305
F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16

```
> confint(model_ols, "kids", level = 0.95)
           2.5 %      97.5 %
kids -135.3298 -28.32302
```

增加一名小孩對每年旅行的英里數之影響的
95%信賴區間為[-135.3298, -28.32302]

#

(6)



從左圖可看出點隨著收入增加而散得更開的漏斗狀現象，此暗示了異質變異的存在。

#

(c) $H_0: \sigma_{\text{前}}^2 = \sigma_{\text{後}}^2$ vs $H_1: \sigma_{\text{前}}^2 \neq \sigma_{\text{後}}^2$

Goldfeld-Quandt test

```
data: model_ols
GQ = 3.1041, df1 = 86, df2 = 86, p-value = 1.64e-07
alternative hypothesis: variance increases from segment 1 to 2
```

$\therefore P\text{-value} = 1.64 \times 10^{-7} < 0.05 \therefore \text{Reject } H_0 \text{ at } \alpha = 0.05$

There is enough evidence to say that $\sigma_{\text{前}}^2 \neq \sigma_{\text{後}}^2$.

The result shows the model is heteroskedasticity.

#

(d)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-391.5480	142.6548	-2.7447	0.0066190	**
income	14.2013	1.9389	7.3246	6.083e-12	***
age	15.7409	3.9657	3.9692	0.0001011	***
kids	-81.8264	29.1544	-2.8067	0.0055112	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


```
> cat("Robust 95% CI for KIDS:", ci_lower, "to", ci_upper, "\n")  
Robust 95% CI for KIDS: -139.323 to -24.32986
```

發現係數b4沒變，但穩健標準誤通常會變大，導致信賴區間變寬，表示在異質變異下，傳統OLS會低估真實的不確定性。

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(e)

常規 GLS (WLS) 的 kids 95% 信賴區間:

```
> print(ci_gls_regular)  
      2.5 %      97.5 %  
kids -119.8945 -33.71808
```

穩健 GLS (Robust WLS) 的 kids 95% 信賴區間:

```
> cat(ci_gls_robust_lower, "to", ci_gls_robust_upper, "\n")  
-121.4134 to -32.19919
```

透過GLS修正異質變異後得到的kids 95%信賴區間為 [-119.8945, -33.71808]，相較於(a)小題與(d)小題，GLS的區間範圍最窄，證實了 GLS 在處理已知形式的異質變異時具備最高效率，而傳統與穩健 GLS 的結果高度一致，顯示模型對異質變異形式的設定（正比於收入平方）是合理的，其最終結果亦支持小孩數量增加會顯著減少家庭旅遊里程的假說。

#